

Aegis Series-7 HCS

AI Intern – Research & Development

Architecture Design Report — Evidence-First Knowledge Ingestion

Objective

Build a provenance-preserving knowledge layer over a heterogeneous, partially contradictory documentation package and expose a query interface that answers with evidence, version scope, conflicts and explicit gaps.

Architecture

The pipeline uses format-specific ingestion, normalization/entity resolution, atomic provenance-bearing facts, explicit version scope, evidence-first retrieval, and a 23-question evaluation harness.

Knowledge Representation

Each fact stores subject, predicate, object, source, location, trust, method and optional applicability/status. This prevents the 180-bar and 200-bar values from being silently overwritten.

Entity Resolution

HPU/Hydraulic Unit/HP unit are normalized together. P.S.04-A is mapped to PS-04A from the diagnostics note. PS-04 and PS-04A remain distinct physical components. PS-04 and PS-40 remain distinct because PS-40 is in the coolant loop.

Deterministic vs Model-Based

Deterministic parsers are used for PDFs, HTML, XLSX, DOCX, JSON and PPTX; OCR is fallback for scans/images. Version rules, aliases, trust and provenance are deterministic. A TF-IDF retrieval fallback is included for arbitrary questions; no external LLM is required.

Visual Documents

OCR recovers labels, but not reliable line semantics. The hydraulic schematic's control/signal topology is therefore captured in a reviewed diagram-facts sidecar. This is explicitly documented rather than pretending OCR understood the drawing.

Conflict Handling

Controlled manuals and ECNs govern current procedures and changes. Superseded documents remain for historical queries. Low-trust field notes are preserved but cannot silently override controlled sources. Configuration values retain firmware applicability.

Deliberate Uncertainty

Questions 19–22 are not determinable from the package. Question 23 is also not established as compatible with 3-phase 400V: the package documents a 480V incoming disconnect but gives no 400V compatibility statement.

Evaluation

The supplied 23-question benchmark is evaluated quantitatively. The harness reports exact benchmark answer accuracy, mean evidence recall, and unknown-handling accuracy.

Limitations / Next Steps

Production improvements could add OCR confidence/crops, graph storage, dense embeddings + reranking, constrained LLM synthesis, contradiction detection, and a web UI with clickable evidence. The structured fact layer should remain the source of truth.

Metric	Result
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Evaluation questions	23
Exact benchmark answer accuracy	100.0%
Mean evidence recall	100.0%
Unknown-handling accuracy	100.0%

These are benchmark results for the supplied evaluation set, not a claim of general LLM accuracy. Unknown handling is intentionally rewarded when the package does not establish a fact.